



Examination	University	Institute	Year	CPI/%
Graduation	IIT Bombay	IIT Bombay	2028	7.5
Intermediate	HSC	JNV Bagalur	2024	93.00
Matriculation	SSC	JNV Gajnur	2022	94.20

SCHOLASTIC ACHIEVEMENTS

- Earned admission to **Jawahar Navodaya Vidyalaya (JNV)** through a highly competitive district-wide entrance examination **JNVST - 2017**, securing one of the 80 merit-based seats among thousands of applicants [2017]
- Awarded a merit-based scholarship by the **Dakshana Foundation**, selected through a highly competitive national screening process recognizing high-potential students across India for outstanding academic excellence. [2022]

PROFESSIONAL EXPERIENCE

Robotics and Automation Intern | NILATECH [Jun '26 – Jul '26]
Design and Development of a 6-DOF Robotic Manipulator System with Motion Control and Actuation Integration

- Designed and developed a **6 Degrees of Freedom (6-DOF) robotic manipulator from scratch**, including CAD modelling, actuator selection, 3-D mechanical design, assembly, and a completely functional prototype development
- Integrated the **electromechanical system**, incorporating NEMA 11, 17 and 23 motors, motor drivers, and control electronics for multi-axis robotic motion with a precise joint actuation, coordination, repeatable movement accuracy
- Developed an **Arduino Mega 2560-based control unit** with embedded firmware enabling joint-wise angular input control and synchronized actuator movement through real-time motor command generation and a precise execution

Applied Machine Learning Intern | Strategic ERP [May '25 – June '25]
Conducted by Robert Bosch Centre for Cyber-Physical Systems (RBCCPS). IISc Bangalore

- Developed a **Personal Health Assistant Chatbot** by leveraging contemporary **machine learning principles** for personalized user guidance within the research environment at IIT Kharagpur under the guidance of Strategic ERP
- Engineered a **Retrieval - Augmented Generation (RAG) pipeline** to power the chatbot, enabling it to provide accurate, context-aware responses by leveraging a specialized knowledge base of medical text articles from PubMed
- Formally recognized by project leadership for **outstanding diligence and dedication**, receiving positive feedback for technical contributions and fostering a **collaborative team environment, leading to high project satisfaction**

KEY PROJECTS

Electrically Assisted Turbocharger | Team TorqueX [Jul '25 – Nov '25]
student led team dedicated for innovating sustainable mobility

- As a key team member, drove the comprehensive design and analysis of an Electrically Assisted Turbocharger, this project was focused on completely eliminating engine turbo lag for vehicle performance and reduction of emissions
- Informed the project's mechanical design by conducting rigorous, in-depth foundational research, involving analyzing existing turbocharger and their technologies to identify and establish a critical performance benchmarks effectively
- Successfully modeling the powertrain system in MATLAB Simulink for simulation, to accurately predict real-world dynamic behaviour, validate all critical designs, and develop the foundational control logic for the electrical assistance

High-Temperature Applications of Inconel Alloys [Aug '25 – Nov '25]
Guide: Prof. Pradeep Dixit - Department of Mechanical Engineering | Course Project - ME221

- As part of a five-person team, researched the properties and the **applications of Inconel alloys in gas turbines**
- Distilled technical research into an engaging seminar, professionally delivered via Presentation and educational video
- Personally **presented the key advantages of Inconel Alloys**, including its superior strength and thermal stability

IMU Controlled Vehicle | XLR8 [Aug '24 – Sept '24]
Electronics and Robotics Club, Institute Technical Council

- Developed a controlled robotic vehicle using **IMU (MPU6050)** and **Raspberry Pi Pico** for intuitive navigation
- Programmed the Raspberry Pi Pico in **Embedded C**, integrating IMU data processing with **L298N** motor control
- Implemented a basic filtering and calibration of IMU data to improve stability of control during obstacle navigation
- Optimized the system through iterative debugging and integration, ensuring smooth and highly responsive operation.

Simulation and Analysis of Watt’s Linkage Mechanism

[May ’25 – Jul ’25]

Guide: Prof. P . Sheshu - Department of Mechanical Engineering | Course Project - ME444

Developed a **multi-body dynamic model** of Watt’s linkage in **Adams** to simulate piston motion in a steam engine

Designed a **14-joint mechanism (revolute, translational, fixed)** with integrated flywheel for a realistic motion

Applied time-dependent **SFORCE input** to model the steam pressure and analyzed piston displacement and velocity

Validated smooth conversion of **reciprocating motion to continuous rotation**, ensuring an error-free simulation

Wireless Drone Controller Development

[Aug ’24 – Nov ’24]

Guide: Prof. Joseph John - Makerspace| Course Project - MS101

Executed full embedded systems for a wireless drone controller, including **PCB design, soldering, and assembly**

Designed and prototyped an joystick case using Fusion 360 and fabricated via 3D printing and laser cutting

Implemented ESP32 in Arduino IDE for a dual joystick control, wireless communication, and safety protocols

Methodically **calibrated the drone controller** performance to achieve precise throttle, pitch, roll, and yaw control

Epidemic Spread Prediction | CodeCure AI Hackathon

[Apr ’26]

Finalist at national-wide competition organized by IIT BHU

Built an **XGBoost epidemic forecasting model** to predict disease spread using time-series epidemiological data

Processed **285K+ global epidemic records** across **197 countries and 73 provinces**, combining the daily cases, vaccination progress, ICU occupancy, and mobility trends into a structured time-series dataset for accurate forecaste

Engineered **features** with **Pandas, NumPy, Scikit-learn**, and ML pipelines to improve model prediction accuracy

Developed a **Streamlit Web dashboard** for province-level forecasting with **MAE, RMSE** and **MAPE** evaluation

Solar Plant Feasibility & Output Forecasting | SolarSense

[Jan ’26 - Apr ’26]

Guide: Prof. Neeraj Kumbhakarna - Department of Mechanical Engineering | Course Project - ME228

Developed a **machine learning pipeline** by integrating some real-time and forecasted weather data from a **climate prediction API Keys**, extracting irradiance, temperature, and temporal features to predict solar AC power output

Designed a **two-stage system** integrating real-time data and historical data for feasibility analysis and forecasting

Implemented and evaluated **Logistic Regression and XGBoost** models to capture patterns in solar generation

Engineered features and built a preprocessing pipeline for **34k+ data points**, enabling reliable predictions, feasibility decisions (GO / NO-GO), and estimating potential profit or loss for the specified local installation area accurately

TECHNICAL SKILLS

Languages	Python, C/C++, L ^A T _E X
Software	MSC Adams,Git/GitHub, Docker, MATLAB, Arduino IDE, Anaconda, Autodesk Fusion360, G-code (NC Programming), VS Code, Jupyter Notebook, Google Colab, Canva
ML & Data Science	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, ChromaDB, LangChain, LangGraph

RELEVANT COURSES

Mechanical Engineering	Thermodynamics, Engineering Mechanics, Solid Mechanics, Structural Materials, Fluid Mechanics, Mechanical Processing of Materials, Analysis & Design of Mechanical Systems and Applied Data Science and Machine Learning
Mathematics	Calculus, Linear Algebra and Ordinary Differential Equations
Other Courses	Applied Data Science and Machine Learning, Introduction to Biology, Introduction to Literature, Makerspace 101, Computer Programming and Utilization, Introduction to Management and Design Thinking for Innovation

EXTRA CURRICULAR ACTIVITIES

Sports	Participated in state-level and district-level Chess tournaments, representing the school.
Cultural	Selected for an intensive 'Art in Education Workshop' by Navodaya Vidyalaya Samiti, based on artistic merit, where I developed in fiber Bas-relief sculpture and canvas painting Pursued 50+ hours of Flute training under National Sports Organisation , IIT Bombay
Miscellaneous	Demonstrated strong STEM aptitude through consistent participation SOF Olympiads. As a culmination of rigorous training, demonstrated rapid and advanced mental calculation skills by competing as a participant in the official State Level Abacus Competition. Completed a workshop on core Data Structures and Algorithms, covering implementation of arrays, linked lists, stacks, and queues, organized by IIT Kharagpur and Strategic ERP.